

4.3.6

EE25BTECH11022 - sankeerthan

Question: passing through the points $(3, 4, -7)$ and $(1, -1, 6)$.

solution: let Passing through the points

$$\mathbf{A} = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix} \quad (1)$$

The direction vector of line passing through $\mathbf{A}$ and $\mathbf{B}$ is

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 1 - 3 \\ -1 - 4 \\ 6 - (-7) \end{pmatrix} \quad (2)$$

$$= \begin{pmatrix} -2 \\ -5 \\ 13 \end{pmatrix} \quad (3)$$

The equation of line is

$$\mathbf{r}(t) = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix} + t \begin{pmatrix} -2 \\ -5 \\ 13 \end{pmatrix} \quad (4)$$

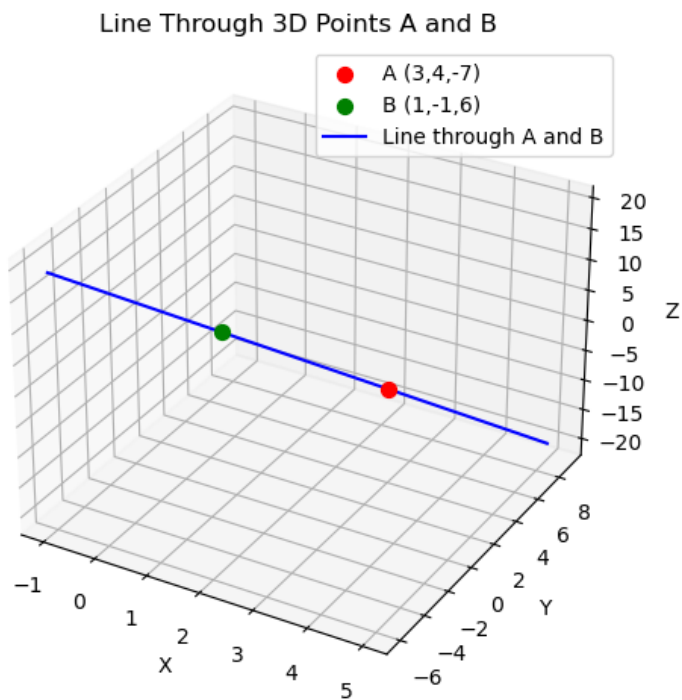


Fig. 0.